

Math100C - WK10

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2026.06.09

1.

Let G be a group of order p^n for a prime p and integer $n > 1$. Prove G has a normal subgroup of order p .

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 Consider the class equation for G :

$$\#G = \#Z(G) + \sum_i \#\mathcal{O}(x_i)$$

The size of the nontrivial orbits (i.e., orbits of elements not in the center) each divide $\#G = p^n$, i.e., $p \mid \#\mathcal{O}(x_i)$ for each i . Suppose G has a trivial center. Then

$$p^n - 1 = \sum_i \#\mathcal{O}(x_i)$$

and p divides the right hand side by the observation above. But clearly p does not divide the left hand side, a contradiction. It must be that G has a nontrivial center; by the order of the group, $p \mid \#Z(G)$. Namely, there is an element of order p in the center; this element generates a normal subgroup of G of order p .

2.

Let K/F be a Galois extension and p any prime number. Show there exists an intermediate field $K/E/F$ such that

- E/F has degree not divisible by p .
- K/E is Galois with degree a power of p .

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 Given a Galois extension K/F , recall that for any intermediate field $K/E/F$, K/E is Galois. If $p \nmid [K : F]$, then just take $E := K$. Assume $[K : F] = p^n a$ for $p \nmid a$ and $n \geq 1$. By Sylow's first theorem, there is a subgroup $H < \text{Gal}(K/F)$ of order p^n . Using the Galois correspondence, one immediately checks that the fixed field of H satisfies the desired properties.

3.

Show that every finite extension of $\mathbb{R}$ other than itself has even degree. Then, together with the previous problems, prove that every finite extension of $\mathbb{C}$ is trivial; in other words, every polynomial over $\mathbb{C}$ splits completely.

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Take any finite, nontrivial extension $K/\mathbb{R}$. By the primitive element theorem, there is an element $\gamma \in K$ such that $K = \mathbb{R}(\gamma)$. $\mathbb{R} \subsetneq K$ shows γ is of degree greater than 1 over $\mathbb{R}$. Since every non-linear odd-degree polynomial over $\mathbb{R}$ has a root, the minimal polynomial for γ must be of even degree. Hence $[K : \mathbb{R}]$ is even.

Suppose there exists a finite, nontrivial extension $K/\mathbb{C}$. Since $\mathbb{C} = \mathbb{R}(i)$ is a degree two extension of $\mathbb{R}$, $K/\mathbb{R}$ is a finite extension. Let L/K be the Galois closure of K with respect to $\mathbb{R}$. Applying the case $p = 2$ to the result of Q2, there is an intermediate field L/E containing $\mathbb{R}$ such that $[L : E] = 2^t$ and $2 \nmid [E : \mathbb{R}]$. As shown above, the only finite extension of $\mathbb{R}$ of degree is the trivial extension; it follows $[L : \mathbb{R}] = 2^t$, where $t > 1$ by assumption on L, K and $\mathbb{C}$.

If $t = 2$, i.e., $[L : \mathbb{R}] = 4$, then namely L is a degree 2 extension of $\mathbb{C}$, i.e., $L = \mathbb{C}(\sqrt{a})$ for some $a \in \mathbb{C}$. But De Moivre's theorem says $\sqrt{a} \in \mathbb{C}$, hence $L = \mathbb{C}$, a contradiction. Hence assume $t > 2$. By Q1, the Galois group $\text{Gal}(L/\mathbb{R})$ of order 2^t has a normal subgroup of order 2, thus yielding a degree 2^{t-1} Galois extension of $\mathbb{R}$. In fact, repeating this argument gives Galois extensions $L_0 = L/L_1/L_2/\dots/L_t = \mathbb{R}$ of degrees $2^t, 2^{t-1}, 2^{t-2}, \dots, 2^0 = 1$, respectively. Now, the L_{t-1} , the degree 2 extension of $\mathbb{R}$, is just $\mathbb{C}$ since every degree two extension of $\mathbb{R}$ is $\mathbb{C}$; $\mathbb{C}$ itself is degree 2 over $\mathbb{R}$, and if $\mathbb{R}(\sqrt{-a})$, $a \in \mathbb{R}_+$, is any degree two extension of $\mathbb{R}$, then $\mathbb{R}(\sqrt{-a})$ contains $\sqrt{-a}\sqrt{1/a} = i$ and therefore $\mathbb{C}$ as well. But then L_{t-2} is a degree 2 extension of $\mathbb{C}$, once again contradicting the fact that $\mathbb{C}$ doesn't have any quadratic extensions.

The conclusion is that there are no nontrivial finite extensions $K/\mathbb{C}$. Namely, this shows every polynomial over $\mathbb{C}$ splits: if an irreducible polynomial over $\mathbb{C}$ has degree greater than 1, then it furnishes a nontrivial finite extension of $\mathbb{C}$, a contradiction.

4.

Prove every abelian group occurs as the Galois group of some Abelian extension of $\mathbb{Q}$.

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I follow a line of reasoning sketched out in Dummit and Foote. Let G be a finite abelian group. By the classification theorem for finitely generated modules over PIDs, namely the PID $\mathbb{Z}$,

$$G \cong (\mathbb{Z}/n_1\mathbb{Z}) \times (\mathbb{Z}/n_2\mathbb{Z}) \times \dots \times (\mathbb{Z}/n_k\mathbb{Z})$$

I will just write Z_{n_i} in place of $\mathbb{Z}/n_i\mathbb{Z}$. By the (true) assumption that there are infinitely many primes congruent to 1 mod any integer m , one can take distinct primes $p_1, \dots, p_k$ such that

$$\begin{aligned} p_1 &\equiv 1 \pmod{n_1} \\ &\vdots \\ p_k &\equiv 1 \pmod{n_k} \end{aligned}$$

Since the primes $p_1, \dots, p_k$ are distinct (namely, p_i, p_j are coprime for $i \neq j$), one can apply the Chinese Remainder Theorem: letting $n = p_1 \cdots p_k$,

$$\mathbb{Z}/n\mathbb{Z} \cong (\mathbb{Z}/p_1\mathbb{Z}) \times \dots \times (\mathbb{Z}/p_k\mathbb{Z})$$

as rings. Of course, this isomorphism of rings induces an isomorphism between the group of units:

$$\begin{aligned} (\mathbb{Z}/n\mathbb{Z})^\times &\cong (\mathbb{Z}/p_1\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_k\mathbb{Z})^\times \\ &\cong Z_{p_1-1} \times \cdots \times Z_{p_k-1} \end{aligned}$$

where the last isomorphism recalls the fact that the multiplicative group of $\mathbb{Z}/p\mathbb{Z}$ for any prime p is cyclic.

Now, by construction, for each $i \in \{1, \dots, k\}$, $n_i \mid p_i - 1$. Recall that abelian groups have a subgroup of any order that divides the order of the group. Hence let $H_i < Z_{p_i-1}$ be a subgroup of order $\frac{p_i-1}{n_i}$; subgroups and quotients of cyclic groups are cyclic, so $Z_{p_i-1}/H_i \cong Z_{n_i}$. This identification gives the isomorphism

$$\begin{aligned} (\mathbb{Z}/n\mathbb{Z})^\times / (H_1 \times \cdots \times H_k) &\cong (Z_{p_1-1}/H_1) \times \cdots \times (Z_{p_k-1}/H_k) \\ &\cong Z_{n_1} \times \cdots \times Z_{n_k} \\ &\cong G \end{aligned}$$

Finally, to realize G as the Galois group of an abelian extension over $\mathbb{Q}$, recall, for any m , $\mathbb{Q}(\zeta_m)/\mathbb{Q}$, where $\zeta_m = e^{2\pi i/m}$ denotes the primitive m th root of unity, is a Galois extension with Galois group $(\mathbb{Z}/m\mathbb{Z})^\times$. Also recall that if a, b are coprime, then $\mathbb{Q}(\zeta_a, \zeta_b) \cong \mathbb{Q}(\zeta_{ab})$. From these facts, one can see

$$\mathbb{Q}(\zeta_{p_1}, \dots, \zeta_{p_k}) \cong \mathbb{Q}(\zeta_n)$$

and this field has Galois group $(\mathbb{Z}/n\mathbb{Z})^\times$ over $\mathbb{Q}$. Furthermore, the Galois group has a (normal) subgroup isomorphic to $H := H_1 \times \cdots \times H_k$, and thus the Galois correspondence says the fixed field $\mathbb{Q}(\zeta_n)^H$ is an (abelian) Galois extension of $\mathbb{Q}$ with Galois group G .